A Tau Core Branch A Higgs-sector module: stabilizer-derived quartic overlap and cohomological lightness protection

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Abstract

We describe a constrained Tau Core Branch A Higgs-sector module in which the observed four-dimensional Higgs is treated as the visible cohomological projection of a tau-profiled parent field. The central reproducible calculation links a minimal $3+2$ stabilizer, the canonical hypercharge direction, a Higgs localization exponent $\nu_D = 3/10$, and a zero-mode quartic overlap $I_4(3/10) = 0.133756$. This value is close to the observed Higgs quartic scale after a small threshold/running correction for a canonical parent quartic. The construction is presented as a candidate mechanism and validation roadmap. It is not a completed Standard Model derivation, not a proof of Tau Core, and not an empirical claim.

1 Scope and Claim Boundary

This manuscript isolates one theoretical module: a possible origin for the Higgs quartic scale and Higgs lightness in a Tau Core projection framework. The paper does not claim that the Standard Model, gravity, or the full Tau Core parent theory has been derived. The result should be read as a reproducible candidate mechanism with explicit proof obligations.

2 Minimal Tau Core Setup

The tau coordinate is treated as an internal projection coordinate rather than an ordinary fifth spacetime dimension. A parent Higgs profile is written as

$$H(x^\mu, x), \tag{1}$$

where x^μ is ordinary four-dimensional spacetime and x is the internal tau/projection coordinate. The observed Higgs is the visible projection

$$\phi(x^\mu) = \Pi_{\text{vis}}[H] = \int h_D(x) H(x^\mu, x)\, dx. \tag{2}$$

Projection-null directions have the form $Q_D^\dagger \Lambda$ and are quotiented out:

$$H \sim H + Q_D^\dagger \Lambda. \tag{3}$$

3 Branch A Stabilizer

Let the parent internal space split as

$$V \simeq \mathbb{C}^5 = C \oplus W, \qquad \dim C = 3, \quad \dim W = 2. \tag{4}$$

The traceless two-block generator is

$$T_\Sigma = \frac{1}{\sqrt{60}}\,\mathrm{diag}(2, 2, 2, -3, -3). \tag{5}$$

Its stabilizer is $S(U(3)\times U(2))$, giving the Standard-Model-compatible non-Abelian stabilizer structure. With

$$Y = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right), \tag{6}$$

the canonically normalized hypercharge generator satisfies $T_\Sigma = -T_Y$.

4 Higgs Exponent and Zero Mode

The Branch A localization rule is treated here as an assumption/theorem-candidate:

$$\nu_i = \frac{3}{5}|Y_i|. \tag{7}$$

For the Higgs doublet, $Y_H = 1/2$, giving

$$\nu_D = \frac{3}{5} \cdot \frac{1}{2} = \frac{3}{10}. \tag{8}$$

The first-order operator is

$$Q_D = \partial_x + \frac{3}{10}\tanh x, \tag{9}$$

and the visible zero mode solves $Q_D h_D = 0$:

$$h_D(x) = \mathcal{N}_D\,\mathrm{sech}^{3/10}\,x. \tag{10}$$

5 Quartic Overlap

For a normalized profile h_ν, the quartic localization functional is

$$I_4(\nu) = \int h_\nu^4 dx = \frac{\Gamma(\nu + 1/2)^2\Gamma(2\nu)}{\sqrt{\pi}\,\Gamma(\nu)^2\Gamma(2\nu + 1/2)}. \tag{11}$$

For $\nu_D = 3/10$ this gives

$$I_4(3/10) = 0.133756. \tag{12}$$

If the parent quartic is $\lambda_\tau(H^\dagger H)^2$, then

$$\lambda_H = \lambda_\tau I_4(3/10). \tag{13}$$

For $\lambda_H \simeq 0.129$, the implied parent quartic is

$$\lambda_\tau \simeq 0.964. \tag{14}$$

This is close enough to a canonical parent value to motivate a threshold/running calculation, but the canonical parent quartic is not derived in this manuscript.

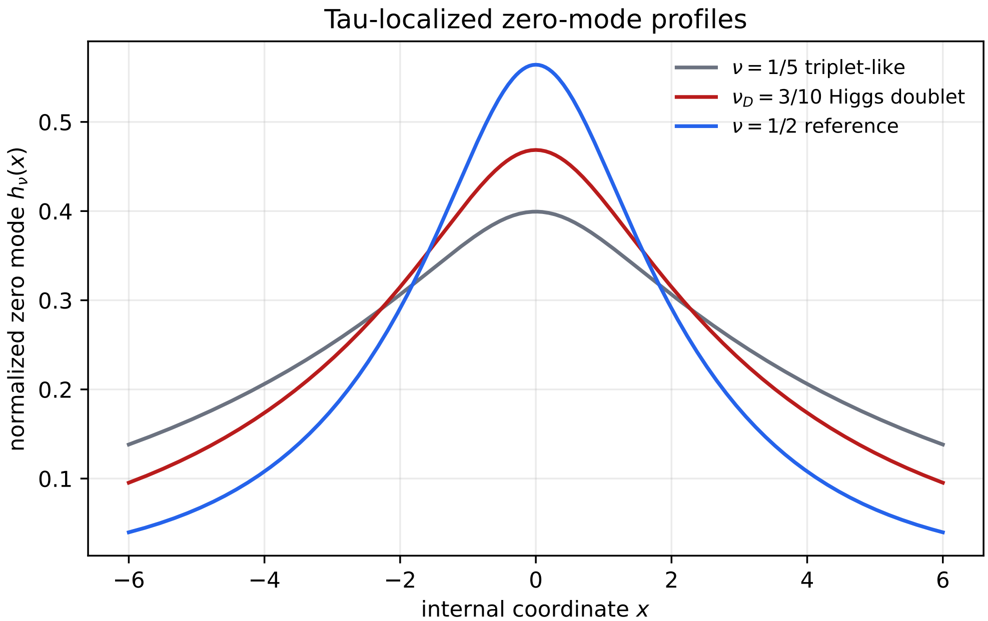

Figure 1: Normalized tau-localized zero-mode profiles. The Higgs doublet exponent $\nu_D = 3/10$ is compared with a triplet-like exponent and a reference exponent.

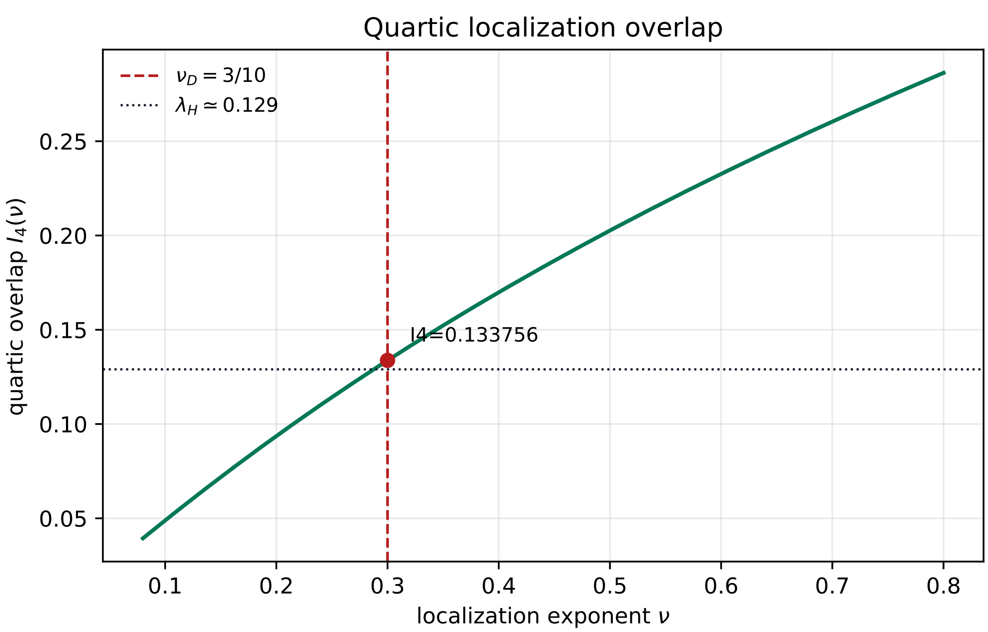

Figure 2: Quartic localization overlap as a function of the tau-localization exponent. The Branch A Higgs value $\nu_D = 3/10$ lands near the observed Higgs quartic scale for an order-one parent quartic.

6 Cohomological Protection

The quotient $H \sim H + Q_D^\dagger \Lambda$ suggests a Projection-BRST implementation,

$$sH = Q_D^\dagger c, \qquad sc = 0, \qquad s\bar{c} = b, \qquad sb = 0. \tag{15}$$

At the exact cohomological point the critical operator is factorized,

$$\mathcal{O}_0 = Q_D^\dagger Q_D, \tag{16}$$

so the visible zero mode is massless. A local parent mass $\int H^\dagger H \, dx$ does not descend to the quotient because it is not invariant under the vertical redundancy. This is the proposed lightness-protection mechanism, but a full anomaly and regulator proof remains open.

7 Controlled Top/Flavor Deformation

The leading cohomology-breaking deformation is written as

$$\mathcal{O}_D = Q_D^\dagger Q_D - \delta_\star W_D, \qquad W_D(x) = \frac{3}{10} \operatorname{sech}^2 x. \tag{17}$$

The zero-mode expectation value gives

$$\int h_D^2 W_D dx = \frac{9}{80}, \tag{18}$$

and therefore

$$\mu_H^2 = \frac{9}{80} \delta_\star M_\tau^2. \tag{19}$$

This section is a roadmap calculation: the top determinant must still be completed before the deformation can be promoted to a derived prediction.

8 Validation Gates

The module would fail if any of the following gates fail:

- the Branch A projection metric rule $\nu_i = 3|Y_i|/5$ cannot be derived;
- the parent quartic is not canonical or requires large tuning;
- the Projection-BRST quotient is anomalous or regulator-dependent;
- a visible Higgs mass is unavoidable at $\delta_\star = 0$;
- the top determinant produces a large mismatch-independent or linear mass term;
- the TeV-sector spectral window is excluded by Higgs coupling or direct-search constraints.

9 Conclusion

The Branch A Higgs module links a $3 + 2$ stabilizer, hypercharge normalization, a $\operatorname{sech}^{3/10}$ visible zero mode, and a quartic overlap close to the observed Higgs quartic scale. The paper establishes a compact, reproducible candidate mechanism. It does not establish the full parent theory. The next paper-grade step is to complete the parent-quartic, Projection-BRST, and top-determinant gates and compare the implied heavy-sector window against collider constraints.

References